Aditya Parikh

adinparikh@gmail.com • +45 91779540 • LinkedIn • GitHub • Google Scholar

Education

Technical University of Denmark (DTU) *Copenhagen, Denmark*
PhD, Department of Applied Mathematics & Computer Science (DTU Compute) *Sept 2025 – Present*
Focus: Responsible AI

Universität Erlangen–Nürnberg (FAU) *Erlangen, Germany*
M.Sc. in ACES (AI & Connectivity) *Oct 2022 – May 2025*
Thesis: Representation Learning for Laparoscopic Imaging

Experience

Doctoral Researcher *Sept 2025 – Present*
DTU Compute (DTU), Kongens Lyngby, DK
- PhD research on 'Responsible AI' – machine learning for medical data, with emphasis on developing and rigorously evaluation on vision and vision-language models, analyzing decoding effects, saliency behavior, bias/fairness, and clinical fidelity for reliable radiology deployment.

Research Assistant (Machine Learning) *Sept 2023 – Aug 2025*
Fraunhofer IAIS, Sankt Augustin, DE
- Built VLM and weak/single-positive learning pipelines for medical imaging (report-to-image generation, CVM classification, CheXpert-style labels), improving accuracy and reducing annotation cost.

Master Thesis Student *Dec 2024 – May 2025*
Machine Learning & Data Analytics Lab (with Fraunhofer IAIS), Erlangen, DE
- Designed encoder–decoder segmentation frameworks with organ-specific decoders and knowledge sharing, improving multi-organ laparoscopic segmentation under class imbalance.

Research Internship (Project Thesis) *Aug 2023 – May 2024*
Machine Learning & Data Analytics (MAD) Lab, Erlangen, DE
- Developed multimodal depression predictive model using facial and physiological signals (data collection → model evaluation) to identify facial experession attributes linked to clinical depression.

Research Assistant (Algorithm Development) *Jan 2023 – Aug 2023*
Fraunhofer IIS, Erlangen, DE
- Developed adaptive filtering methods for non-stationary automotive acoustic echo cancellation, improving early convergence by ∼35% and runtime efficiency by 12%.

Deep Learning Engineer (Intern) *Jul 2022 – Sept 2022*
PixelApps.io, Ahmedabad, IN
- Built a school attendence pipeline achieving 97%+ accuracy across varied lighting/poses & robust to camera angle.

Machine Learning Engineer (Intern) *May 2021 – Dec 2022*
BizAmica Softwares Pvt. Ltd., Pune, IN
- Co-designed an AI-powered document processing system that reduced manual entry time by 10 folds and improved end-to-end workflow reliability for population census data (1950 to 2010) for Govt. of India.

Publications

1. **Measuring What VLMs Don't Say: Validation Metrics Hide Clinical Terminology Erasure in Radiology Report Generation**. **Aditya Parikh**, Aasa Feragen, Sneha Das & Stella Frank. (under review), 2026.
2. **Fair Lung Disease Diagnosis from Chest CT via Gender-Adversarial Attention Multiple Instance Learning**. **Aditya Parikh** & Aasa Feragen. CVPR workshop (PHAROS-AIF-MIH; under review), 2026.
3. **Investigating Label Bias and Representational Sources of Age-Related Disparities in Medical Segmentation**. **Aditya Parikh**, Sneha Das & Aasa Feragen. **Oral**. *Int. Symposium of Biomedical Imaging (ISBI)*, 2026.
4. **Who Does Your Algorithm Fail? Investigating Age and Ethnic Bias in the MAMA-MIA Dataset**. **Aditya Parikh**, Sneha Das & Aasa Feragen. **Oral**. *MedEurIPS (NeurIPS-endorsed workshop)*, 2025.

5. **Leveraging Transfer Learning with Class-Specific Decoders for Laparoscopic Segmentation**. Priya Priya*, **Aditya Parikh***, et al. *IEEE Workshop on Big Data Analytics for Medical Imaging*, 2025.
6. **Representation Learning for Laparoscopic Imaging: A Study of Architectural Design and Multi-Level Knowledge Transfer**. **Aditya Parikh**, Sai Jitin Jami, et al. *Master Thesis (grade 1.3), OpenFAU*, 2025.
7. **Learning from a Few Shots: Data-efficient Cervical Vertebral Maturation Assessment**. Helen Schneider*, **Aditya Parikh***, et al. *MIDL*, 2025.
8. **Effective Disjoint Representational Learning for Anatomical Segmentation**. Priya Priya*, **Aditya Parikh***, et al. *MIDL*, 2025.
9. **First Investigation of Deep Learning for Intraoperative Gauze Segmentation in Minimally Invasive Abdominal Surgery**. Priya Priya*, **Aditya Parikh**, et al. *IEEE DSAA*, 2025.
10. **The Potential of Acknowledging the Unknown: Single Positive Multi-label Learning in Medical Image Processing**. Helen Schneider, Priya Priya, **Aditya Parikh**, et al. *MIDL*, 2024.
11. **Informed Deep Abstaining Classifier: Investigating noise-robust training for diagnostic decision support systems**. Helen Schneider, Sebastian Nowak, **Aditya Parikh**, et al. *ICONIP*, 2024.
12. **Mandibular Advancement Devices: Impact on Occlusal Dynamics and Tooth Alignment Modifications**. Aylin Uyaner, Helen Schneider, **Aditya Parikh**, et al. *Dentistry Journal (MDPI)*, 2024.
13. **Exploring Facial Biomarkers for Depression through Temporal Analysis of Action Units**. **Aditya Parikh**, Misha Sadegi, Bjorn Eskofier. *Project Thesis*, 2024.
14. **Information Extraction from Unstructured Data using Augmented-AI and Computer Vision**. **Aditya Parikh**. *IJERT*, 2022.

Skills

Technical Skills	Python*, C++, LaTeX, MATLAB, SQL
Libraries & Packages	PyTorch*, NumPy*, Pandas*, Scikit*, Matplotlib*, TensorBoard, HuggingFace, PyTorch-Lightning, W&B
Software Tools	Git*, Bash, Linux, Docker, VSCode*, Jupyter, ONNX, Azure ML
Cloud Platforms	AWS, Google Cloud Platform (GCP)
High-Performance Tools	SLURM, MPI
Research Interests	Representation learning, multimodal models, robustness, bias and evaluation of foundation models

Achievements & Responsibilities

- **Teaching Assistant (Responsible AI):** Supported 70+ students on fairness, accountability, and evaluation.
- **Peer review:** Reviewer for MedEurIPS workshop, ICAPSS, and *Medical Image Analysis (MedIA)*.
- **Hackathons:** Winner (Smart Farming, 2024); Winner (Kicklehack, 2024); Runner-Up (Tensor Tournament, 2023); Runner-Up (EBEC Case Study, 2023).
- **Competitions:** Kaggle Expert (Notebook rank #1500); MachineHack global rank 310 (top 4%).
- **Certificates:** Selected coursework and certifications.
- **Open-source:** Selected GitHub projects.